Hints: Polynomials – 5. Integer coefficients, divisibility and irreducibility

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

(a)

Hint 1. If some $p(a_i) = 0$, take $a = a_i$. Otherwise use $u \equiv v \pmod m \Rightarrow p(u) \equiv p(v) \pmod m$: if $a \equiv a_i \pmod{p(a_i)}$ then $p(a_i) \mid p(a)$. The moduli $p(a_i)$ need not be coprime, so work prime by prime.

Hint 2. For each prime q dividing $\prod p(a_i)$, let q^e be the largest power of q dividing some $p(a_{i(q)})$, and require $a \equiv a_{i(q)} \pmod{q^e}$. These congruences have coprime moduli, so the Chinese remainder theorem gives a, and then $q^e \mid p(a)$ for every q.

(b)

Hint 1. Not always. Look for an example where two values share a prime factor q but q^2 never divides $p(a)$.

Hint 2. $p(x) = x^2 + 1$, $a_1 = -1$, $a_2 = 1$: the product is 4, but $a^2 + 1 \not\equiv 0 \pmod 4$.

Problem 2

Hint 1. First reduce to rational coefficients. If $f = P/Q$ and $f(n_i) = m_i \in \mathbb{Z}$ for infinitely many n_i, the conditions $P(n_i) = m_i Q(n_i)$ form a linear system with rational coefficients for the coefficients of P, Q.

Hint 2. The system has a nonzero real solution, hence a nonzero rational one (P_1, Q_1); then $PQ_1 - P_1Q$ vanishes at all n_i, so $f = P_1/Q_1$. Thus we may assume $P, Q \in \mathbb{Z}[x]$.

Hint 3. Divide: $P = SQ + R$ with $\deg R < \deg Q$ and $S \in \mathbb{Q}[x]$. With D a common denominator of the coefficients of S, the numbers $Df(n_i) - DS(n_i) = D\,R(n_i)/Q(n_i)$ are integers tending to 0, so they vanish for large i. Hence $R = 0$.

Problem 3

Hint 1. Take $a = b = c = n$ to get $n^2 \mid P(n)$ for all $n \neq 0$; deduce $P(0) = 0$ and that the coefficient of x vanishes.

Hint 2. Compare (a, b, c) with $(-a, b, c)$: $a^2 + b^2 + c^2$ divides $P(a) - P(-a)$ for all b, c, and $b^2 + c^2$ can be arbitrarily large. So P is even, $P(x) = R(x^2)$.

Hint 3. Now $c^2 \equiv -(a^2 + b^2)$ gives $P(a) + P(b) + P(c) \equiv R(a^2) + R(b^2) + R(-a^2 - b^2) \pmod{a^2 + b^2 + c^2}$; letting c grow shows $R(a^2) + R(b^2) + R(-a^2 - b^2) = 0$ for all integers a, b, hence $R(x) + R(y) + R(-x - y) \equiv 0$ as polynomials.

Hint 4. With $R(0) = 0$ this says R is odd and additive, so $R(x) = kx$. Answer: $P(x) = kx^2$ with $k \in \mathbb{Z}$ (and these work).

Problem 4

Hint 1. $P(x) = \pm x$ works. If $P = A^2 B$ with A non-constant, then $|A(p)| > 1$ for some prime p; so assume P is squarefree in $\mathbb{Q}[x]$, i.e. $\operatorname{Res}(P, P') \neq 0$. Also write $P = x^k Q$ with $Q(0) \neq 0$: $k \geqslant 2$ fails at once, and $Q = c$ constant forces $c = \pm 1$.

Hint 2. If Q is non-constant, its values have infinitely many prime divisors (Schur). Pick a prime $q \mid Q(m)$ with $q \nmid Q(0)\operatorname{Res}(Q, Q')$; then $q \nmid Q'(m)$ and $q \nmid m$.

Hint 3. By Hensel's lemma there is $r \equiv m \pmod q$ with $q^2 \mid Q(r)$, and $\gcd(r, q) = 1$. By Dirichlet's theorem there is a prime $p \equiv r \pmod{q^2}$, and $q^2 \mid Q(p) \mid P(p)$. Answer: $P(x) = \pm x$.

Problem 5

Hint 1. You only need k distinct small divisors of $f(n)$ below n. Find k different primes that all divide $f(n)$ for suitable n.

Hint 2. By Schur's theorem infinitely many primes divide some value of f; choose $p_1 < \cdots < p_k$ and r_i with $p_i \mid f(r_i)$.

Hint 3. By the Chinese remainder theorem every $n \equiv r_i \pmod{p_i}$ (all i) has $p_1 \cdots p_k \mid f(n)$; for such $n > p_k$ with $f(n) \neq 0$ take $d_i = p_i$ and $d_{k+1} = f(n)/(p_1 \cdots p_k)$.

Problem 6

Hint 1. Let $t = \sqrt[3]{5}$ and $\alpha = t + t^2$. Find the minimal polynomial of α over $\mathbb{Q}$.

Hint 2. $\alpha^3 = 30 + 15\alpha$, and $x^3 - 15x - 30$ is Eisenstein at 5. Dividing P by this monic integer polynomial, $P(\alpha) = u + v\alpha + w\alpha^2$ with $u, v, w \in \mathbb{Z}$.

Hint 3. With $\alpha^2 = 10 + 5t + t^2$ and the basis $1, t, t^2$: the coefficients of t^2 and t give $v + w = 0$ and $v + 5w = 1$, so $w = \frac{1}{4}$, contradiction.

Problem 7

Hint 1. $p(x) - x$ vanishes at $3 + \sqrt{5}$, so it is divisible by its minimal polynomial $x^2 - 6x + 4$. Since this is monic, $p(x) = x + (x^2 - 6x + 4)q(x)$ with $q \in \mathbb{Z}[x]$.

Hint 2. At $x = 3 - \sqrt{7}$: $x^2 - 6x + 4 = (x - 3)^2 - 5 = 2$, so $p(3 - \sqrt{7}) = 3 - \sqrt{7} + 2q(3 - \sqrt{7})$ with $q(3 - \sqrt{7}) \in \mathbb{Z}[\sqrt{7}]$.

(a)

Hint 1. You would need $2q(3 - \sqrt{7}) = 1$, impossible in $\mathbb{Z}[\sqrt{7}]$. Answer: no.

(b)

Hint 1. You need $q(3 - \sqrt{7}) = 1$: take $q = 1$, i.e. $p(x) = x^2 - 5x + 4$. Answer: yes.

Problem 8

Hint 1. Linear polynomials $ax+b$ with $a \neq 0$ rational work, constants do not. For $\deg f = n \geqslant 2$, find an irrational x with $f(x) \in \mathbb{Z}$.

Hint 2. Rational roots of $f(x) = q$ ($q \in \mathbb{Z}$) have denominators dividing a fixed integer L (rational root theorem applied to $D(f - q)$, D a common denominator).

Hint 3. Take M with f increasing on $[M, \infty)$. For each integer q between $f(M)$ and $f(X)$ there is a root of $f = q$ in $[M, X]$; there are about X^n such q but only $L(X - M) + 1$ rationals with denominator dividing L in $[M, X]$. For large X some root is irrational. Answer: $f(x) = ax + b$, $a, b \in \mathbb{Q}$, $a \neq 0$.

Problem 9

Hint 1. Repunits are $R_m = \frac{10^m - 1}{9}$, i.e. $9R_m + 1 = 10^m$. Conjugate: $g(y) = 9f(\frac{y-1}{9}) + 1$ maps every 10^m ($m \geqslant 1$) to a power of 10.

Hint 2. If $g(x) = cx^d + \ldots$, then $g(10^m)/10^{md} \to c$, and $g(10^m)/10^{md}$ is an integer power of 10. So the exponent is eventually constant, $c = 10^e$, and $g(x) = 10^e x^d$ (agreement at infinitely many points).

Hint 3. The condition that $g(10) = 10^{e+d}$ is a power of 10 with exponent $\geqslant 1$ gives $e + d \geqslant 1$. Answer: $f(x) = \frac{10^e(9x+1)^d - 1}{9}$ with integers $d \geqslant 0$, $e \geqslant 1 - d$.

Problem 10

Hint 1. Compute $p^{(j)}(k)$ for $p(x) = \sum a_i x^i$: it is $j! \sum_i \binom{i}{j} a_i k^{i-j}$. So it is always divisible by $j!$, and $p = x^j$ shows that $j!$ is the best possible.

Hint 2. So you need the least j with $2016 = 2^5 \cdot 3^2 \cdot 7 \mid j!$. Answer: $j = 8$ ($7! = 5040$ is not divisible by 2^5).